INTRODUCTION TO ESA'S FLUORESCENCE EXPLORER (FLEX)

JULIANE BENDIG, IRENEUSZ KLEPPERT & UWE RASCHER

10TH VIRTUAL EEBIOMASS WORKSHOP: "SHINING LIGHT FROM PHOTOSYNTHESIS: REMOTE SENSING OF FLUORESCENCE" - 28.09.2023

http://flex-mission.eu

FLEX – ESA‘S PHOTOSYNTHESE MISSION

- **Address key scientific challenges**
- **Demonstrate breakthrough technology**

- **Improve our understanding of the global carbon cycle**
- **Measure photosynthetic activity from space**

https://earth.esa.int/eogateway/documents/20142/3007097/Next-Earth-Explorers-infographic.pdf

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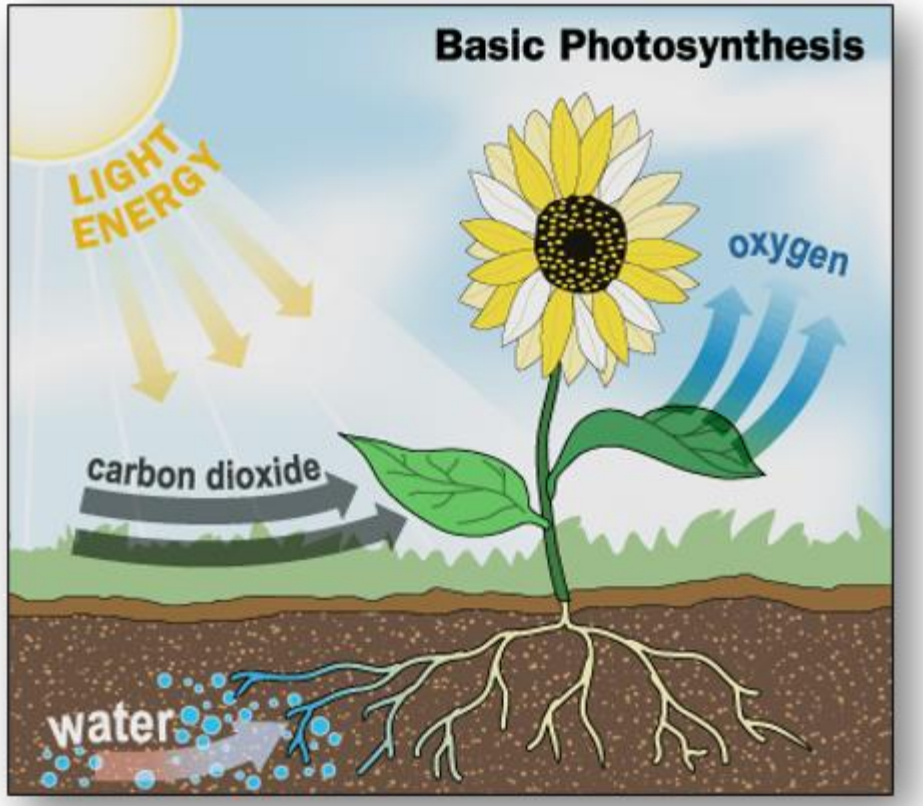
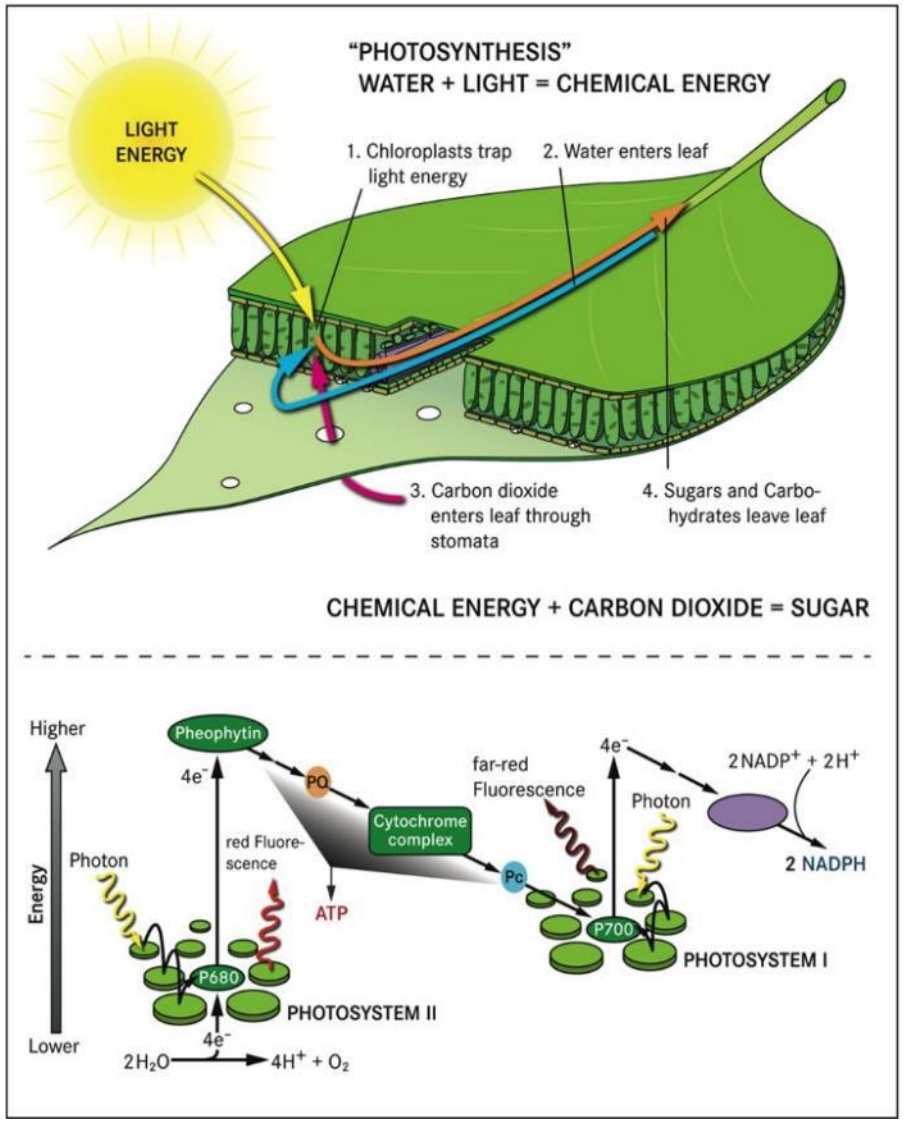
INTERACTION OF LIGHT AND PLANTS



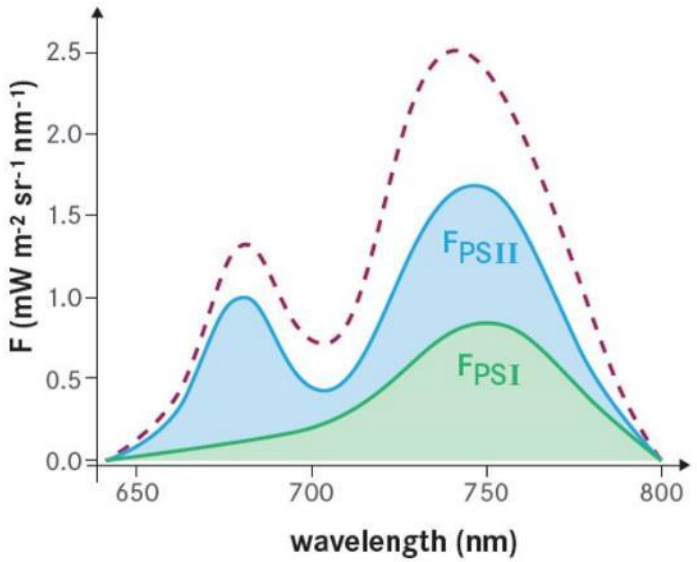
Abou-Hussein 2012,
Journal of Applied Sciences Research 8(8):4359-4383

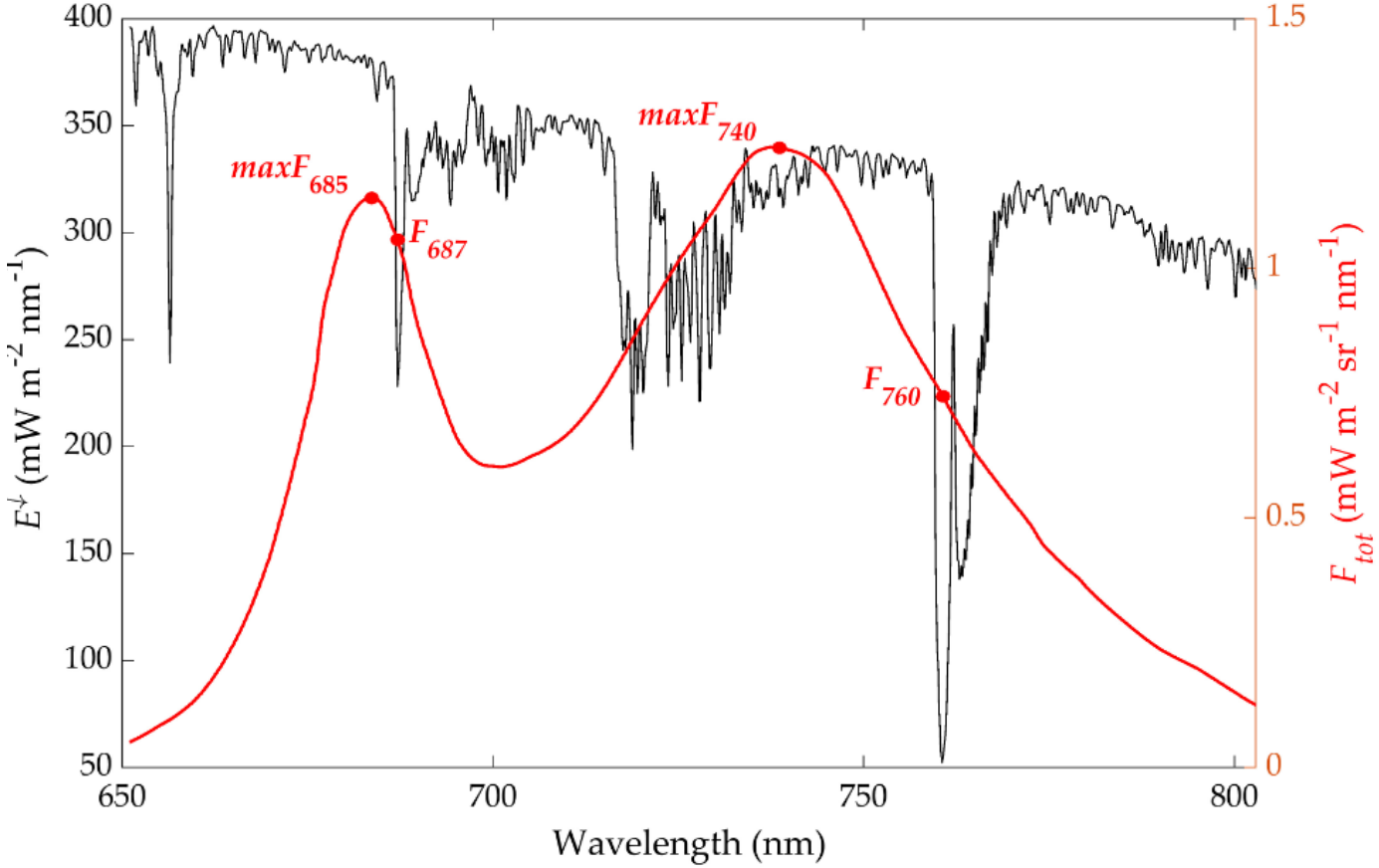


ESA 2015,
Report for Mission Selection: FLEX

MEASURING PHOTOSYNTHESIS

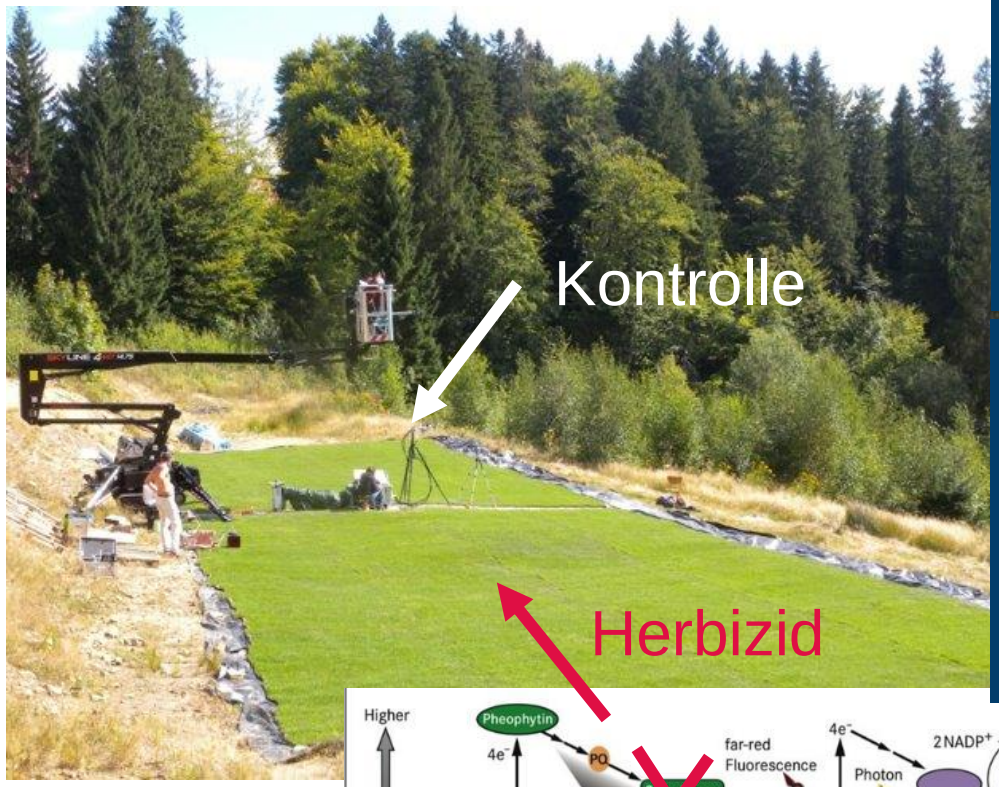
With remote sensing methods

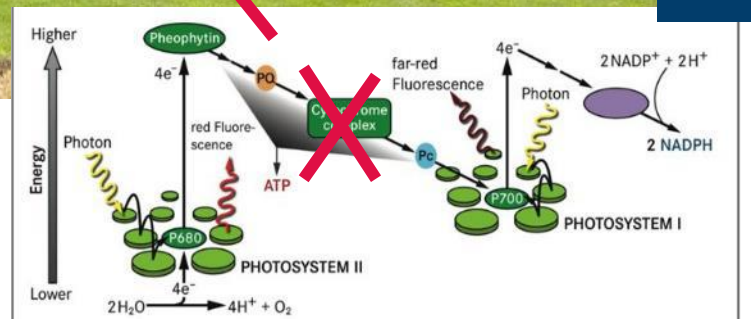


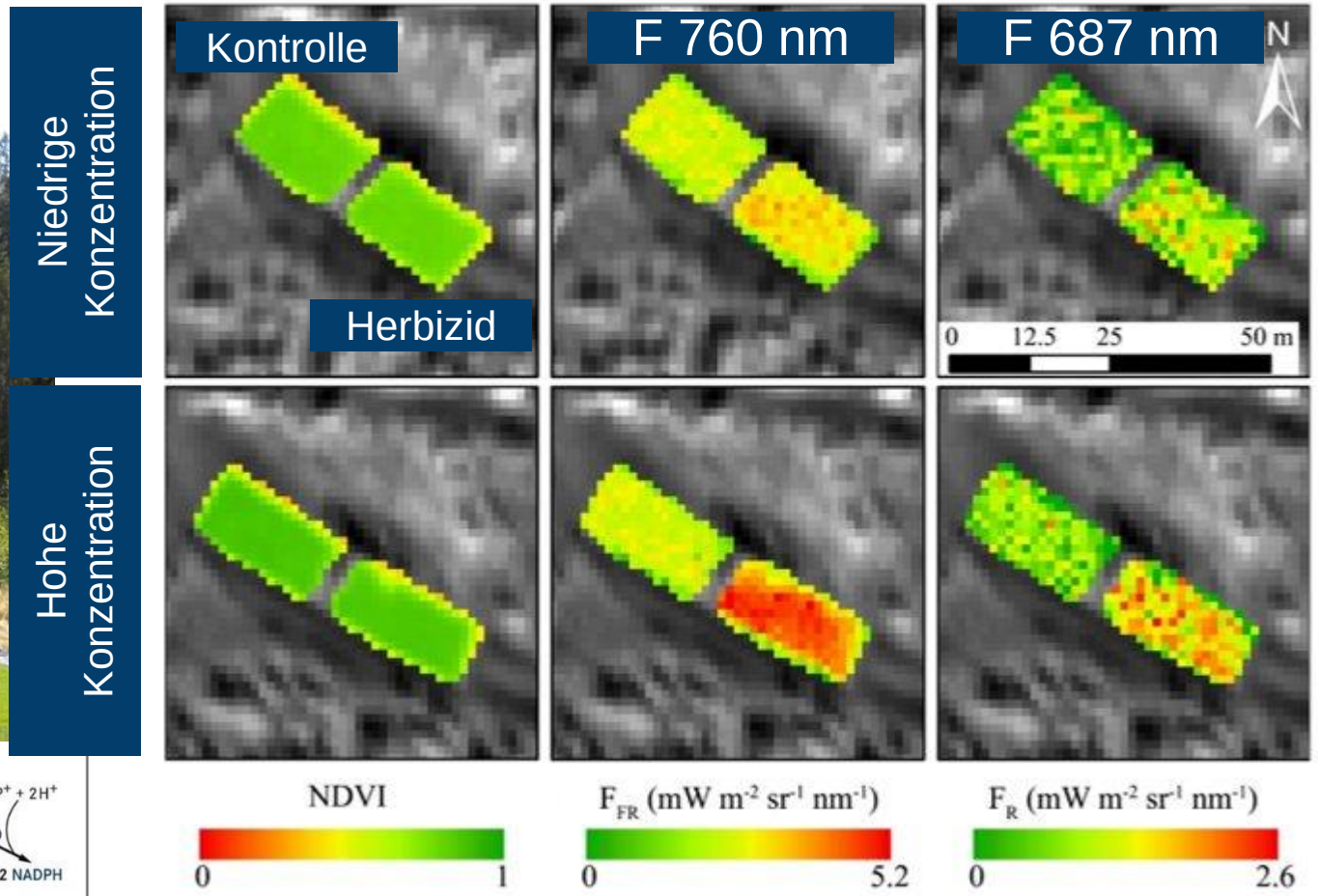


ESA 2015, Report for Mission Selection: FLEX
Cendrero-Mateo et al. (2019) Remote Sensing 11

PROOF OF CONCEPT







Rossini et al. 2015, Geosphysical research letters, 10.1002/2014GL062943

FLEX – OVERVIEW

Sensor
Fluorescence Imaging Spectrometer (FLORIS) with varying spectral resolution in the visible and near infrared wavelengths

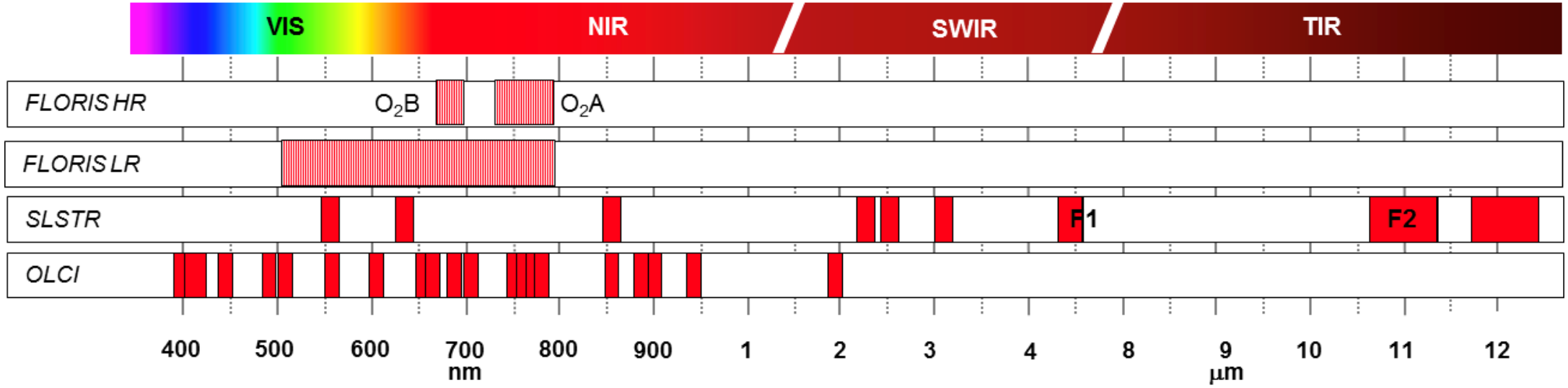
Orbit
814 km height above ground, 10 am local acquisition time

Special condition
Tandem orbit with Copernicus Sentinel-3 for optical (SWIR) & thermal data

Facts
Start 2026
Mission duration 3.5 years
Repetition rate 10-27 days
Pixel size 300 x 300 m

https://www.esa.int/Applications/Observing_the_Earth/FutureEO/FLEX

FLORIS, OLCI, SLSTR



FLORIS (300 m SSD*)
- Fluorescence
- Photochemical reflectance changes
- Atmospheric correction
- True reflectance

*Spatial sampling distance

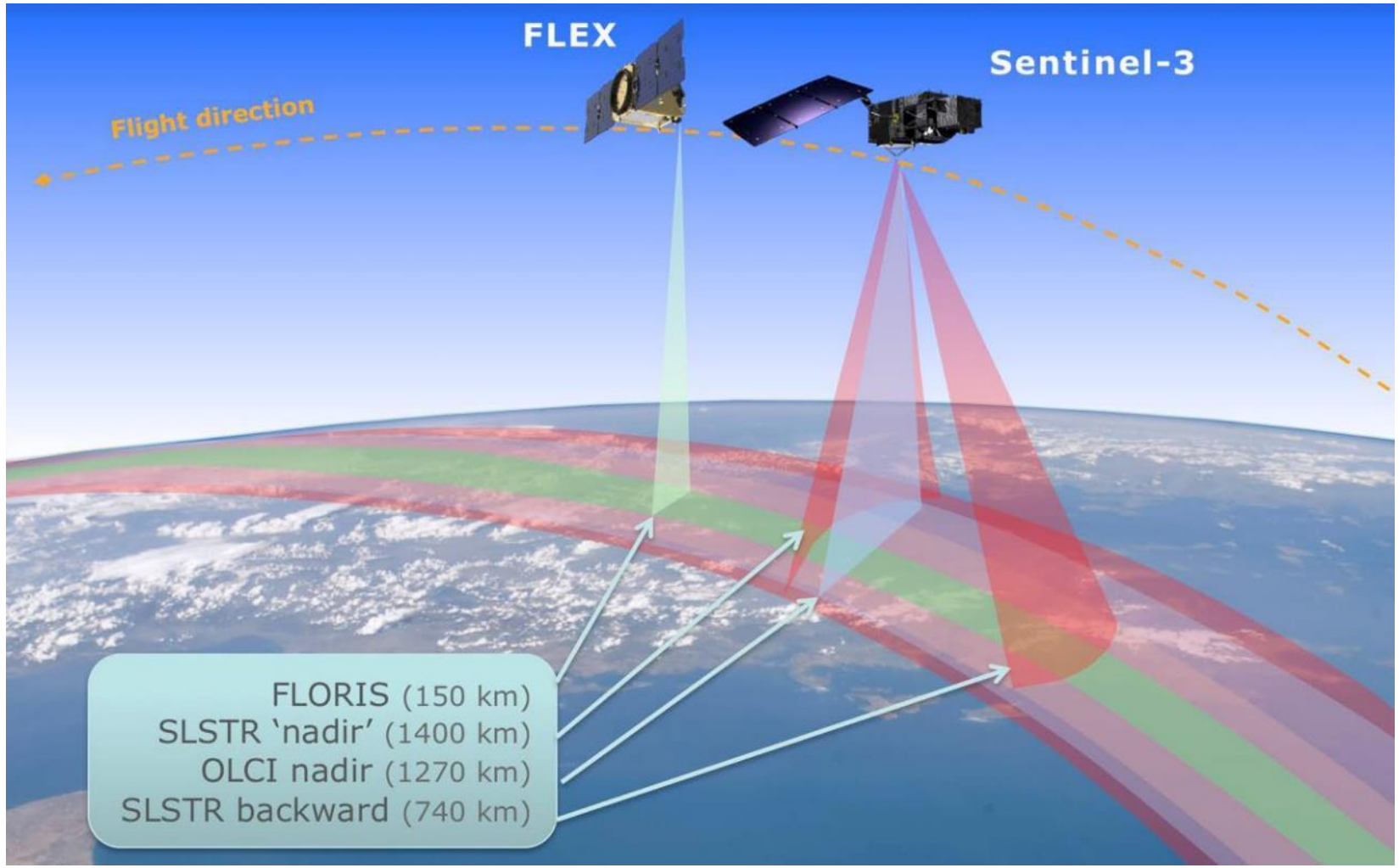
S-3 OLCI-4 (300 m SSD)
- Aerosols & water vapour
- Biophysical variables (Chlorophyll, LAI, ...)
- Reflectance
- Cross calibration
- Context information

OLCI = Ocean and Land Colour Instrument
SLSTR = Sea-land surface temperature

SLSTR (500m –1 km SSD)
- Aerosols
- Surface reflectance
- Land surface temperature
- Clouds (cirrus)

Source: Matthias Drusch, ESA

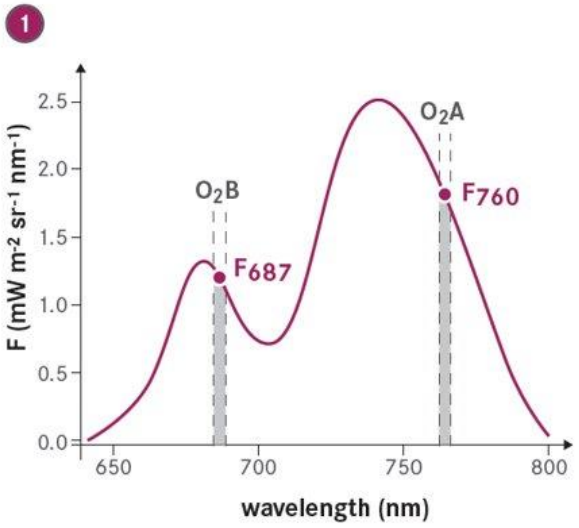
FLEX – SENTINEL 3 - TANDEM

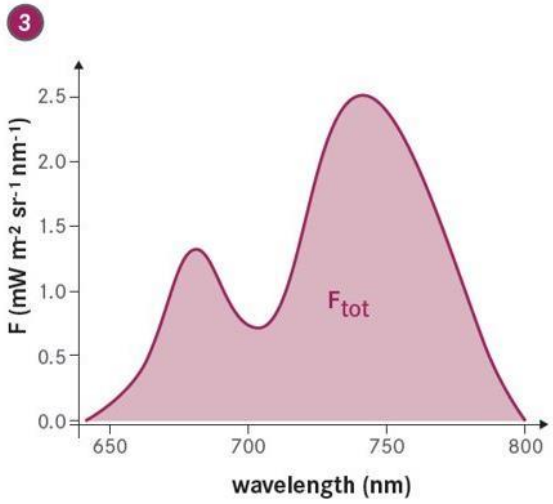


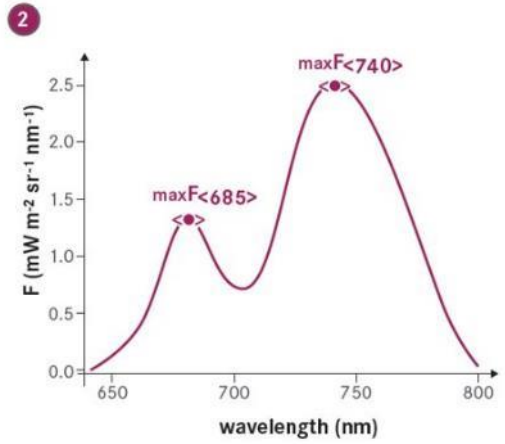
https://www.eoportal.org/satellite-missions/flex#tandem-mission-with-sentinel-3

DATA PRODUCTS

- **Level-0: raw data**
- **Level-1b: radiometrically calibrated**
- **Level-2a: georectified, tiled like Sentinel-3**
- **Level-2b, c, d: FLORIS, OLCI & SLSTR geophysical parameters (incl. fluorescence)**
- **Level-x:**
  - **Re-absorption corrected fluorescence**
  - **Electron transport rate**
  - **CO_2 uptake**
  - **Gross primary production (GPP)**
  - **Stress indicators …**







uncertainty requirements
0.2 mW m^{-2} sr^{-1} nm^{-1}
5 nm for max F peaks

2018 FLEX mission requirements
https://download.esa.int/docs/FLEX/FLEX_MRD_v3.0_20180605_issued.pdf

FLEX – MISSION STATUS

Updates from FLEX Workshop 19-21 September - Frascati

- Launch date updated to 2026 due to launch vehicle availability
- Critical design review scheduled for June 2024

- Scientific readiness assessment completed:

  FLEX positively evaluated – remaining challenges:

  - Readiness of data products definition
  - Performance assessment through end-to-end simulations
  - Data dissimination strategy
  - Cal/val strategy

FLEX –COMMUNITY DISCUSSION

Updates from FLEX Workshop 19-21 September - Frascati

- More SIF measurements over open waters & coastal zones
- Make FLEX simulated test datasets available to users to speed up algorithm development
- Make existing ground & airborne campaign datasets accessible
- In-situ data needs uncertainties, protocols & other meta data to be useful for the community
- Data assimilation / data fusion / SIF as essential climate variable (ECV) → discussion ongoing → watch terrestrial community carbon assimilation system (TCCAS) (T. Kaminski et al.)

Source: Twitter/X ESA Earth Observation @ESA_EO

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THE GERMAN FLEX PROJECT OFFICE

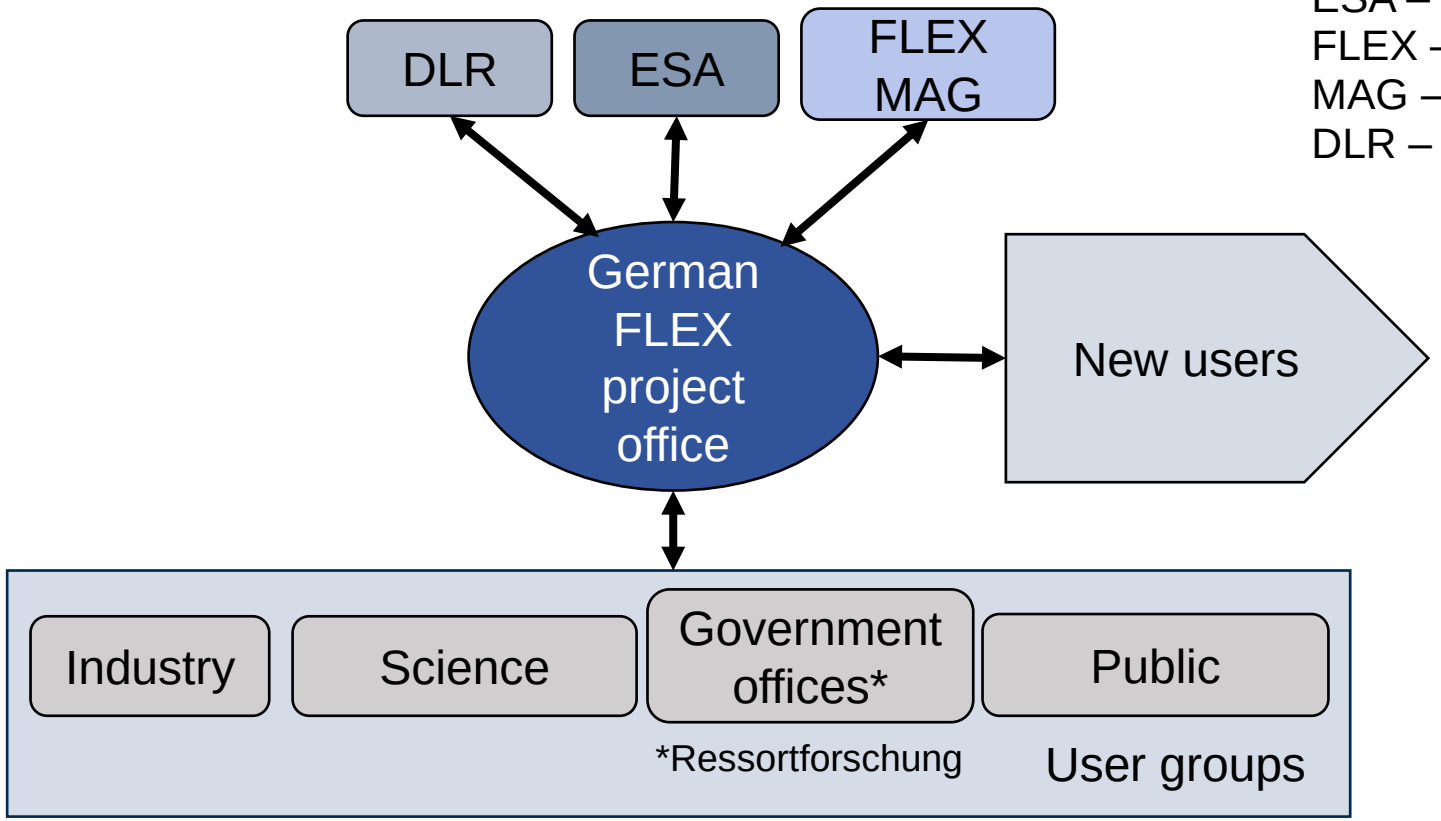
ESA – European Space Agency
FLEX – FLuorescence Explorer
MAG – Mission Advisory Group
DLR – German Aerospace Center

Supported by:

Federal Ministry for Economic Affairs and Climate Action

on the basis of a decision by the German Bundestag

DLR



We connect satellite/information providers and users since April 2023

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WHAT WE DO

Mission

The FLuorescence EXplorer (FLEX) mission will collect global chlorophyll fluorescence data from vegetation, providing clues to photosynthetic activity and plant health. This information is essential to better understand and characterize the global carbon cycle. This understanding can support developments in agriculture and contribute to food security.

That is why FLEX was selected as the eighth Earth Explorer Mission of the European Space Agency (ESA) Living Planet program in 2015 (mission selection report). FLEX will fly in tandem with the Copernicus Sentinel-3 mission so that data from the OLCI and SLSTR instruments can contribute to the interpretation of the FLEX data.

Data about FLEX:

- Start: 2026
- Launcher: Vega-C
- Mission duration: 3.5 years
- Size: 1.5 x 1.2 x 1.2 m (LxWxH), 5.2 m wide after unfolding the solar panels
- Satellite mass: 460 kg (of which 140 kg instruments, 30 kg fuel)
- Orbit: sunsynchronous, at 814 km, inclination 98.64°
- Repetition rate: 27 Tage

Sources:

FLEX Mission Overview
more information

FLEX Satellite Design
more information

Ride into space on Vega-C secured for FLEX and Altus
more information

Inform about FLEX

In German & English

Publikationen

Thematisch sortierte Publikationsliste zu FLEX und Fluoreszenz

→ Publikationen

Berichte

Übersicht von Berichten zur FLEX Mission wie Mission selection report

→ Berichte

Kampagnen

Flugzeugkampagnen im Zusammenhang mit der Vorbereitung der FLEX Mission

→ Kampagnen

Tools

Strahlungstransfermodelle, spektrale Datenbanken, Orbitsimulation

→ Tools

Datenpakete

SIF Testdaten von FloX & HyPlant (in Vorbereitung)

→ Anfrage

Projekte

Forschungsprojekte im Zusammenhang mit FLEX und Fluoreszenz

2023-28 FLEXINEL

Provide auxillary data

- FLEX-Satelliten-Mission – Technische und programmatische Umsetzung +
- Übersichtsartikel über sonnen-induzierte Fluoreszenz +
- SIF-retrieval Methoden & Ansätze +
- Bodengestützte Messungen der sonnen-induzierten Fluoreszenz +
- Aktivitäten im Zusammenhang mit cal/val +
- Unoccupied Airborne Vehicle (UAV)-basierte Ansätze zur Messung von sonnen-induzierter Fluoreszenz +
- Flugzeuggestützte Sensoren zur Messung von sonnen-induzierter Fluoreszenz +
- Sonnen-induzierte Fluoreszenz Messungen von existierenden Satellitenplatformen +
- Verstehen der funktionellen Grundlagen der Sonnen-induzierten Fluoreszenz – Photosynthese-Modelle +
- Sonnen-induzierte Fluoreszenz und Brutto-Primärproduktivität (GPP) Modellierung +
- Sonnen-induzierte Fluoreszenz als früher Stressindikator +
- Sonnen-induzierte Fluoreszenz im Wasserkreislauf +
- Sonnen-induzierte Fluoreszenz für die Wasserforschung +
- Weitere Anwendungen von sonnen-induzierter Fluoreszenz - z.B. Biodervisität, Stickstoffstatus, Altersstruktur +

Publications by topic

WHAT WE DO

- What do users need FLEX for?
- Which data products will they use?
- Which data products would they like to have?

Language: English - English | Change the language

User survey FLourescence EXplorer (FLEX) - ESA's 8th Earth Explorer

Welcome to our FLEX user survey!

The FLuorescence EXplorer (FLEX) is a scientific satellite of the European Space Agency (ESA) that will collect global data of chlorophyll fluorescence from plants starting in 2025 (info FLEX). These data will provide insights into the photosynthesis of plants from space. From these observations, we can for example investigate plant health or plant stress.

With this survey, we at the FLEX project office would like to ask potential users about their previous experience and perception of FLEX. We would also like to find out what specific requirements you might have for FLEX data. Even if you have never heard of FLEX, your responses are of great value to us.

By participating, you will help us greatly!

Best regards

Juliane Bendig, Ireneusz Kleppert & Uwe Rascher

There are 21 questions in this survey.

This survey is anonymous.

The record of your survey responses does not contain any identifying information about you, unless a specific survey question explicitly asked for it.

If you used an identifying access code to access this survey, please rest assured that this code will not be stored together with your responses. It is managed in a separate database and will only be updated to indicate whether you did (or did not) complete this survey. There is no way of matching identification access codes with survey responses.

Next

Survey - Identify user needs

18.01.2024 FLEX-Nutzerworkshop

1. FLEX-Nutzerworkshop

Forschungszentrum Jülich | Wilhelm-Johnen-Str. 52428 Jülich, Deutschland

18.01.2024

Beim FLEX-Nutzerworkshop sind alle eingeladen darüber zu diskutieren, v

Wir wollen die Interessen der deutschen Nut
erarbeiten. Wir möchten Sie, die FLEX Nutzer

→ Zum Workshop anmelden!

Workshop – Discuss data products

TAKE HOME MESSAGES

FLEX

- Launch 2026
- 300 x 300 m pixel global SIF
- ~every month
- Data products, dissimination & cal/val under development

German FLEX project office

- satellite mission status
- SIF research overview
- Link & support users in Germany & worldwide

FLEX user workshop
18.01.24
at Forschungszentrum Jülich

FLEX user survey

http://flex-mission.eu
info@flex-mission.eu

FLuorescence EXplorer (FLEX) project office | Projektbüro

@FLEXofficeDE

@info-FLEX

29.09.2023 FLEX user seminar

Reconstruction of the full SIF spectral profile using emulation

29.09. 2 pm

Miguel Morata Dolz (Universitat de València) explains how to derive the entire fluorescence emission spectrum with emulation.

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